

YUVRAJ SINGH

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TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (EC2, IAM, VPC), Terraform, Docker, Kubernetes
- **CI/CD & Observability:** GitHub Actions, Jenkins, Prometheus, Grafana
- **Development:** React.js, FastAPI, Flask, REST APIs
- **Languages:** Java, Python, Bash
- **Databases & Tools:** MySQL, Supabase, Git/GitHub, Linux
- **Core CS:** DSA, Operating Systems, Computer Networks, DBMS

PROJECTS

Automated Cloud Application & Observability Stack [View Repo](#)

- Engineered a robust Python/FastAPI REST API backend, achieving high uptime and improved response times through custom Prometheus metrics.
- Provisioned and version-controlled AWS infrastructure (EC2) as code (IaC) using Terraform for repeatable, automated deployments.
- Significantly reduced deployment time by moving from a manual setup process to a fully automated CI/CD pipeline.
- Orchestrated a multi-container environment using Docker Compose, integrating application, Prometheus, and Grafana services with inter-container networking and persistent volumes, greatly reducing manual setup effort, and simulated real-world failure scenarios via chaos engineering (stress testing).

Production Kubernetes Reliability & Observability Platform [View Repo](#)

- Containerized and deployed a FastAPI microservice on a Kubernetes (kind) cluster using Deployments, Services, ConfigMaps, and liveness probes to ensure high service availability and automatic recovery from failures.
- Implemented custom application observability using Prometheus client libraries, exposing core metrics (request count, latency histogram) scraped at regular intervals via a Kubernetes ServiceMonitor.
- Deployed Prometheus and Grafana using Helm, built real-time dashboards for request rate and p95 latency, and performed root cause analysis using application metrics during simulated workload spikes.

Flask Application on AWS using Terraform & Jenkins [View Repo](#)

- Engineered a fully automatic CI/CD pipeline to build and deploy a containerized 2-tier Flask + MySQL application on AWS EC2 using Jenkins, Docker, and Docker Compose, triggered on every Git push.
- Provisioned and managed AWS infrastructure (EC2, security groups, SSH key pairs) as code using Terraform, replacing manual console setup with a single reproducible terraform apply/destroy workflow.
- Automates server bootstrapping (Docker, Jenkins, Java) via EC2 user-data scripts, cutting environment setup time from a multi-step manual process to a fully hands-off deployment.

CERTIFICATIONS

- [AWS Cloud Practitioner Certified | Issued: June 2026](#)
- [eJPTv2 Certified | Issued: Feb 2026](#)
- [The Bits and Bytes of Networking — Google \(Coursera\)](#)

EXPERIENCE

Intern — India Space Lab

- Built and maintained company's website while working on national space club making and Python tutorials.

EDUCATION

VIT Bhopal University — B.Tech in Computer Science and Engineering, CGPA: 8.36

Sant Atulanand Convent School — Senior Secondary Education, 83.6%

Sant Atulanand Residential Academy — Secondary Education, 90.8%

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